First, as we have seen, for all planar graphs and all graphs whose crossing number is smaller than 4, we can use the straight-line drawing without any loss; not all the other graphs with $cr(G) \geq 4$, moreover, have a worse $rcr(G)$ than their $cr(G)$.

Turns out, for practical purposes we are often more interested in drawing planar or almost-planar graphs, because flow-charts, PERT diagrams, workflows etc… are usually fairly sparse graphs, which in turn are less likely to have a high crossing number: after all, from Euler's invariants we know that planar graphs must be sparse (since the number of edges is linear in the number of vertices).

Second, even if curve-line embeddings can lead to fewer crossing points, it doesn't mean that it's easy to find one; as a matter of fact, these drawings are much harder to find with an algorithm, because we have to optimize many more parameters: besides the position of the vertices, we also have to find the ideal curves that reduce the number of intersections. This will add at least one or two parameters per edge, if Bézier's quadratic or cubic curves are used, making the search space much larger and harder to search: the number of parameters, in fact, goes from $0(V)$ to $0(V+E)$, which in turn is $0(V^2)$ in the worst case.

Third, using curves requires a larger computational power: the algorithm to check if two segments intersect is significantly easier than checking if two curves do (and it can be made even easier with some restrictions on the position of the vertices). This means that even computing the